Government of Canada / Gouvernement du Canada

V-24—Relevant local standards of volume

Category: Volume

Issue date: 2011-06-01

Effective date: 2011-07-01

Revision number: 2

Supersedes: 1

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Information bulletin—Limits of error for refuellers and retail motor fuel dispensers

1.0 Scope

This bulletin applies to the local standards which may be used when calibrating or certifying volumetric measuring devices. In the event that new technology or circumstances warrant a deviation from the normal approach, inspectors should consult with their regional specialist to determine what relevant local standard will be appropriate in each such case.

2.0 Purpose

The purpose of this bulletin is to define the relevant local standards for particular types, sizes and classes of volumetric device. It is also intended to detail general requirements for volumetric standards and any special conditions of use, where applicable.

3.0 References

Section 25 of the Weights and Measures Act prohibits an inspector from certifying any device for use in trade, without comparing it to a relevant local standard.

4.0 Definitions

Local Standard
(Étalon locale)
Any standard that has been designated by the Minister under section 13 of the Weights and Measures Act.

Pass

(Passage)

Terms of pipe proving standards, is the movement of a pipe prover's displacer from one switch to the other. Except for provers that incorporate pulse interpolation, a single pass must produce at least 10,000 unaltered pulses while the displacer passes through the calibrated section of the prover.

Run

(Essai)

Terms of Small volume Prover (SVP) and a uni-directional pipe proving standards, is a pre-defined as a number of passes of the pipe prover's displacer. For a bi-directional pipe prover, a run consists of one left to right pass and one, right to left pass. In the case of a bi-directional pipe prover, the run must produce at least 20,000 pulses.

5.0 General requirements for local standards

All local standards must meet the design requirements defined by the Measurement Canada manual, "Guidelines for the Calibration and Certification of Volumetric Standards", or as defined by the Measurement Canada Volumetric Engineer.

All local standards must be certified within the time frame stated in section 56 of the Weights and Measures Regulations.

Each local standard must be accompanied by:

- A Certificate of Designation, signed by a person delegated to do so, by the authority of the Minister
- An Inspection Certificate annotated in accordance to Measurement Canada's bulletin V-22 "Sealing requirements for liquid volumetric

proving standards" to identify the reference conditions of the calibration, the essential construction data used in calculation for the calibration of the standard and the number and location of all inspection seals.

Each local standard must be used in accordance with any and all applicable conditions stated on the certificate.

The internal coatings and seals on any volumetric prover must be compatible with the product being tested. If there are any questions regarding this, consult the regional volumetric specialist.

Local standards of volume must employ some means to indicate to the operator what product was last contained therein, to reduce the possibility of product contamination. This can take the form of a product identification tag conspicuously located on the prover to notify the operator of the product last contained.

In order to be considered a valid local standard, an electronic thermometer must be monitored by conducting an ice point test, at minimum once in the previous 30 days before use or whenever the standard's integrity or accuracy is suspect. The results of an ice point test must be recorded throughout the standard's certification period.

6.0 Relevant local standard against which volumetric devices may be certified

6.1 Dispensers for motor fuel

With a flow rate less than 91 lpm - Section 265 of the regulations, which defines the limit of error (LOE) for these devices, does not stipulate any particular known quantity or time frame for the test delivery. Convention

has been to use 20 litre measures but larger measures or provers can be used. Acceptable local standards include both open and closed systems, namely:

- provers or measures; top or bottom filled,
- pipe provers; and
- reference meter provers.

All provers or measures that are used to certify gasoline dispensers must employ recognized means to minimize product vaporization. This does not apply to diesel dispensers or factory tests of dispensers using solvents.

A list of standards that have been evaluated and recognized can be found on Measurement Canada's website.

6.2 High speed refuellers

With a flow rate greater than 90 lpm, used to deliver motor fuel - Section 265 of the regulations, which defines the LOE for these devices, does not stipulate any particular known quantity or time frame for the test delivery. Convention has been to use 250 litre provers, however, smaller provers may be used, but they must be a sufficient size to contain a quantity equal to or greater than one minute's flow at maximum operating rate.

High speed refuellers may also be tested with reference meter provers that are certified to the applicable operating range of the high speed refueller.

6.3 Large capacity PD and turbine meters

Stationary or truck mounted, used to deliver petroleum products - these meters can be tested with a suitably-sized open neck prover or with a suitably-sized pipe prover.

When using open neck provers, they must be of a sufficient size to contain a quantity equal to or greater than one minute's flow at maximum operating rate. In accordance with section 266 of the regulations, provers used with meters 65 mm (2.5 inches) in diameter or smaller must have a capacity between 225 litres and 900 litres. For meters which are 75 mm (3 inches) in diameter or larger, the prover volume must be greater than or equal to 1350 litres.

When using uni-directional or bi-directional pipe provers without acceptable means of pulse interpolation, they shall receive 10,000 or more unaltered pulses during each pass. An unaltered pulse is defined as a pulse generated from an actual event. This can be an opening and or closing of a switch in the case of a mechanical switch; the opening and or closing of a window in the case of an optical switch; or the rising intersect point and or declining intersect point with a threshold for an electromagnetic switch type pulser. Pulses cannot be multiplied by electronic means.

To prevent excessive displacer by-pass or switch "bounce" when using uni-directional or bi-directional pipe provers, the velocity of the displacer should not exceed 1.5 m/s for bi-directional provers (ball type); or, 3 m/s for uni-directional provers; or, 1.0 m/s for bi-directional piston type provers. If it is determined that the displacer velocity is in excess of these limits, the regional volumetric specialist should be consulted to determine if the prover can be used.

6.3.1 Small volume pipe provers

Pipe provers which generate less than 10,000 pulses per pass can be used, provided pulse interpolation is employed. The velocity of the displacer shall not exceed the manufacturer's specified limit.

6.4 Turbine and PD meters measuring liquefied gases

These meters may be tested using a pipe prover, a gravimetric prover or a vapour displacement prover.

6.5 Mass flow meters / Coriolis meters

When the unit of trade on these devices is mass, the devices must be tested with a reference scale. If the unit of trade is in terms of volume and the device is only measuring a single product, the device may be tested with a reference scale and an appropriate means to determine density; or, a suitably sized volumetric standard. If the unit of trade is in terms of volume and used to measure more than one product, the device may be tested with a reference scale and an appropriate means to determine density; or, a suitably sized volumetric standard and an appropriate means to determine density.

Adequate means of determining density are certified pycnometer used with an appropriate scale and certified mass standards, or a certified floating instrument (normally a hydrometer). For further information on determining the density of a product, refer to Measurement Canada's Field Inspection Manual (FIM) and the appropriate Standard Test Procedure (STP).

Open neck volumetric standards, must be of a size adequate to contain a volume of product at least equal to five (5) MMQ for the Coriolis meter under test.

Pipe provers may only be used after consultation with the regional volumetric specialist or volumetric engineer, to determine the appropriateness of the standard.

6.6 Heated products such as bunker or asphalt

These meters must be tested with a pipe prover or by suitable gravimetric means.

7.0 Revisions

The purpose of revision 2 was to:

- Clarify the general requirements for volumetric local standards; to remove the background section which no longer applied; to revise section 6.5 which had stipulated a method for determining the volume of product required to test Coriolis meters; to provide the corrected title for the "Guidelines for the Calibration and Certification Procedures for Volumetric Standards" manual under section 5.1; and, make other necessary editorial corrections.
- Reference section 3.0 was added which replaced the background.
- Definitions section 4.0 was added to define local standard, pass and run.
- Section 6.1 was amended to name the types of acceptable provers and sections 6.1.1 was added to require all types of gasoline dispensers be inspected with appropriate means to minimize vapour loss during a test of the meter as well as section 6.1.2 informing that a list of recognized standards is available.
- Section 6.2 was amended to revise the application of Weights and Measures regulation from 266 to 265 as the applicable LOE for high speed refuellers. Added section 6.2.1 to include reference meter provers as test standards which may be used to test high speed refuellers.

- Added reference to the FIM and STPs in 6.5.1 for further information on determining the density of product required to test Coriolis meters.

The purpose of revision 1 was to change the criteria when large open provers must be used; to add criteria regarding proving meters used with heated products and to correct some minor errors.

Date modified:
2017-05-26